

Credit Card Balance Dataset

Comprehensive Analysis Report

Multi-Perspective Analysis: Business | Product | Data Quality | Data Science

Home Credit Default Risk — credit_card_balance.csv

NUS — Institute for Financial Risk Management (IFRM)

Table of Contents

- 1. Executive Summary
- 2. Dataset Overview & Structure
- 3. Business Analyst View — Real-World Challenges
- 4. Product Analyst View — Usage & Behavioral Patterns
- 5. Data Analyst View — Data Quality & Structural Issues
- 6. Data Scientist View — Predictive Feature Engineering
- 7. Temporal Trend Analysis
- 8. Cross-Dataset Linkage & Value Assessment
- 9. Recommended Feature Engineering Pipeline
- 10. Conclusions & Strategic Recommendations

1. Executive Summary

The **credit_card_balance.csv** dataset contains **3,840,312 monthly snapshots** across **104,307 credit card accounts** belonging to **103,558 unique applicants**, spanning 96 months of history. It is one of the most information-rich supplementary tables in the Home Credit Default Risk competition.

This report analyzes the dataset from four professional perspectives — Business, Product, Data Quality, and Data Science — to extract actionable insights for default risk prediction.

Metric	Value	Significance
Total Records	3,840,312	Monthly snapshots per account
Unique Accounts	104,307	One account per applicant (mostly)
Coverage of Train	28.3% (86,905)	Available for ~1 in 3.5 applicants
Time Span	96 months	8 years of behavioral history
Top Signal	Utilization >80%	16.2–25.5% default rate vs 5.6% base
Key Risk	Over-limit: 8.7%	25.5% default rate — 3x baseline
DPD Prevalence	4.0% of records	Rare but with survivorship bias

2. Dataset Overview & Structure

The dataset has a **panel structure** — each row represents one month of one credit card account. Accounts are identified by **SK_ID_PREV** and linked to applicants via **SK_ID_CURR**.

Column	Type	Missing %	Description
SK_ID_PREV / SK_ID_CURR	int	0%	Account & applicant identifiers
MONTHS_BALANCE	int	0%	Month relative to application (-1 = most recent)
AMT_BALANCE	float	0%	Outstanding balance on credit card
AMT_CREDIT_LIMIT_ACTUAL	int	0%	Current credit limit
AMT_DRAWINGS_ATM_CURRENT	float	19.5%	ATM cash withdrawal amount
AMT_DRAWINGS_POS_CURRENT	float	19.5%	POS purchase amount
AMT_DRAWINGS_OTHER_CURRENT	float	19.5%	Other drawing amount
AMT_DRAWINGS_CURRENT	float	0%	Total drawings this month
AMT_INST_MIN_REGULARITY	float	7.9%	Minimum payment due
AMT_PAYMENT_CURRENT	float	20.0%	Actual payment made
AMT_PAYMENT_TOTAL_CURRENT	float	0%	Total payments this month
AMT_RECEIVABLE_PRINCIPAL	float	0%	Principal receivable
AMT_RECIVABLE	float	0%	Total receivable (note: typo in name)
AMT_TOTAL_RECEIVABLE	float	0%	Total receivable (near-duplicate)
CNT_DRAWINGS_* (4 cols)	float	0–19.5%	Count of drawings by channel
CNT_INSTALLMENT_MATURE_CUM	float	7.9%	Cumulative matured installments
NAME_CONTRACT_STATUS	object	0%	Status: Active/Completed/Demand/etc.
SK_DPD	int	0%	Days past due (current)
SK_DPD_DEF	int	0%	Days past due (official definition)

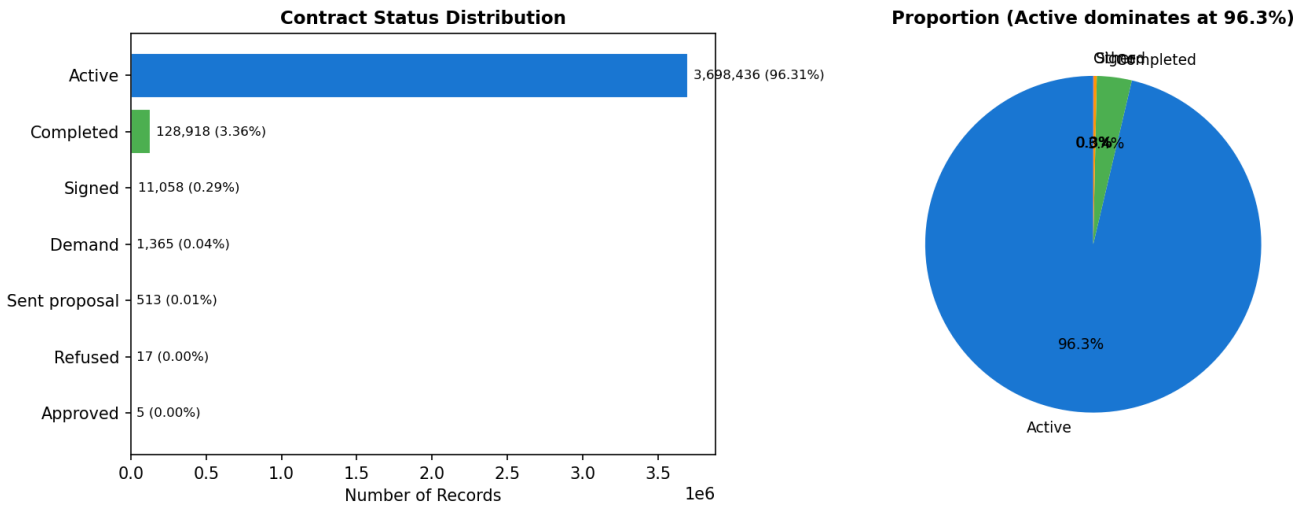
Key Descriptive Statistics:

Feature	Mean	Median	Std Dev	Min	Max
AMT_BALANCE	58,300	0	106,307	-420,250	1,505,902
AMT_CREDIT_LIMIT	153,808	112,500	165,146	0	1,350,000
AMT_PAYMENT_CURRENT	10,281	2,703	36,078	0	4,289,207
SK_DPD	9.3	0	97.5	0	3,260
UTIL_RATIO	0.37	0.01	0.43	-1.80	11.78

3. Business Analyst View — Real-World Challenges

3.1 Contract Status — Extreme Imbalance

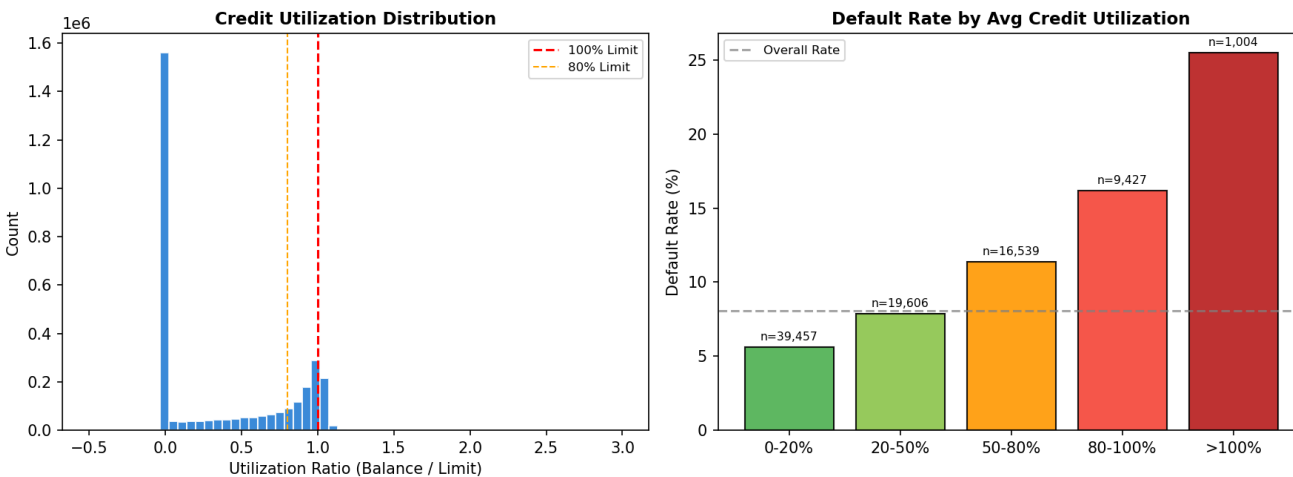
The contract status distribution reveals a heavily skewed portfolio:



96.3% of all monthly records are 'Active', meaning the credit card is in normal use. Only **3.4%** are 'Completed' (fully paid off), and the remaining statuses are extremely rare. The 'Demand' status (1,365 records, 0.04%) represents the lender demanding full repayment — the most severe credit event. Despite its rarity, even one 'Demand' event on a customer's history is a critical risk flag that should trigger immediate review.

3.2 Credit Utilization — The Strongest Risk Signal

Credit utilization (balance / credit limit) is the **single most powerful predictor** in this dataset:



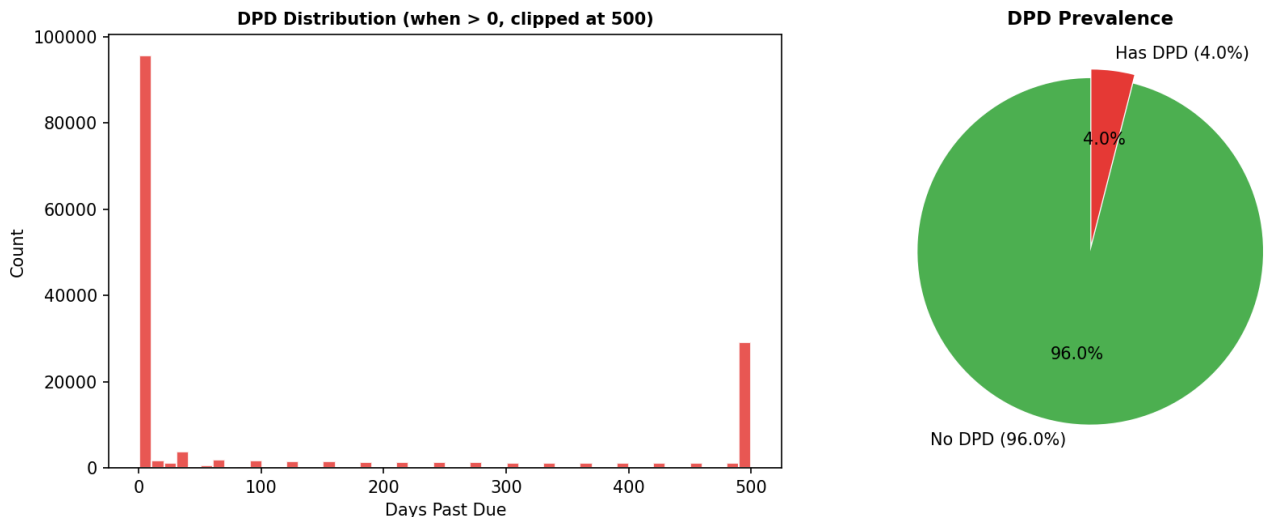
Utilization Band	Default Rate	Count	Risk Level
0–20%	5.6%	39,457	LOW
20–50%	7.9%	19,606	MODERATE
50–80%	11.4%	16,539	ELEVATED
80–100%	16.2%	9,427	HIGH
>100% (over-limit)	25.5%	1,004	CRITICAL

Business Impact: Applicants using >80% of their credit limit default at **2.9–4.5x the rate** of low utilizers. In real-world banking, utilization >80% universally triggers automated limit reviews and risk re-assessment.

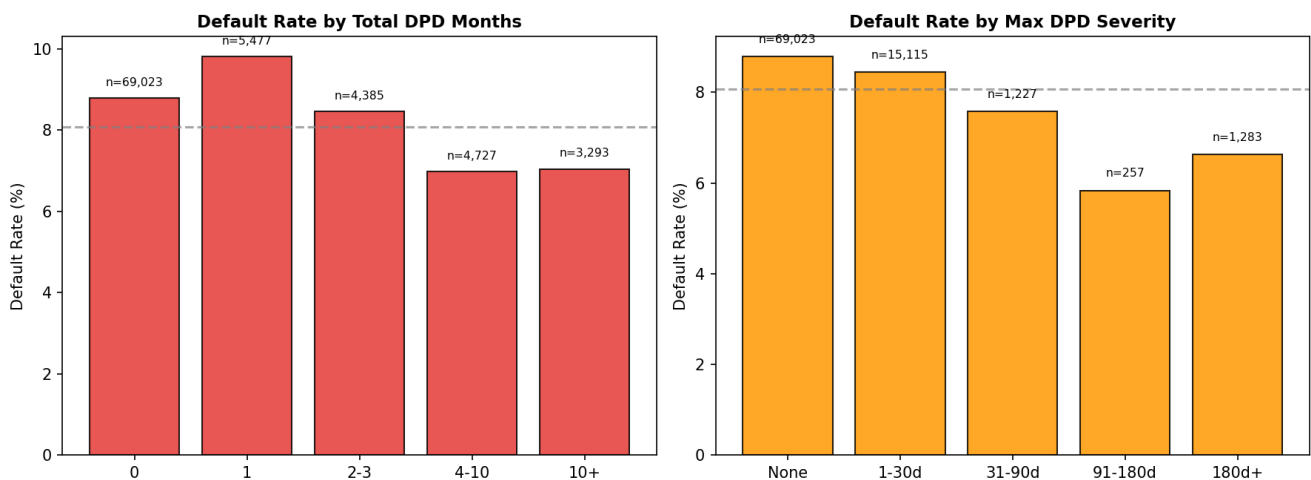
Over-limit borrowing (8.7% of records) indicates either lack of financial discipline or emergency spending — both strong default predictors.

Recommendation: Implement real-time utilization monitoring. Automatically flag applicants whose average utilization exceeds 80% for proactive credit line management.

3.3 DPD Survivorship Bias — A Counterintuitive Finding



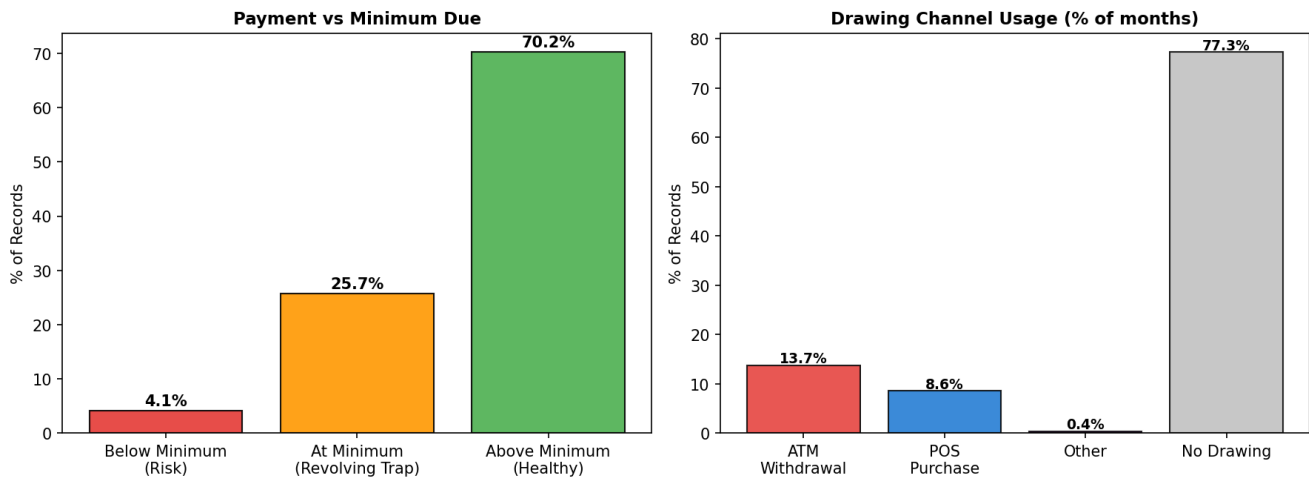
Only **4.0%** of records show any DPD, but the distribution is **bimodal**: median DPD is just 1 day (administrative delays), while the mean is 232.5 days (pulled up by extreme outliers reaching 3,260 days / ~9 years).



Counterintuitive finding: Default rates are actually **lower** for applicants with historical high DPD (91-180 days: 5.8%, 180+: 6.6%) compared to those with no DPD (8.8%). This is a classic **survivorship bias** — customers who recovered from past DPD have proven their ability to catch up. Models must distinguish between *active/recent DPD* vs. *historical recovered DPD*.

4. Product Analyst View — Usage & Behavioral Patterns

4.1 Payment Behavior & Drawing Channels



Payment vs. Minimum Due:

70.2% of payments are above the minimum — indicating a largely healthy portfolio. However:

- **4.1% underpaying** (123,124 records) — these customers are accumulating penalties and interest, compounding debt in a dangerous spiral.
- **25.7% paying exactly the minimum** (776,114 records) — the 'revolving trap'. These customers generate maximum interest revenue for the lender but face elevated default risk. Three or more consecutive minimum-payment months should trigger proactive outreach.

4.2 Drawing Channel Mix

ATM cash withdrawals (13.7% of months) are more common than POS purchases (8.6%). In consumer finance, **cash advances from credit cards are a universally recognized risk signal** — they carry higher interest rates and indicate cash urgency rather than planned spending. The ATM-to-POS drawing ratio is a valuable feature for risk scoring.

4.3 Product Recommendations

1. Flag customers with 3+ consecutive minimum-payment months for restructuring offers (lower rate, fixed payment plan).
2. Implement graduated utilization alerts: 70% (advisory), 85% (warning), 95% (action required).
3. Monitor ATM-heavy users separately — a shift from POS to ATM usage often precedes financial distress.
4. Design a 'balance reduction' incentive program for customers with balance velocity > 0 for 6+ consecutive months.

5. Data Analyst View — Data Quality & Structural Issues

5.1 Missing Data Patterns

Feature Group	Missing %	Likely Cause	Treatment
AMT_DRAWINGS_ATM/POS/OTHER	19.5%	Months before card activation	Fill with 0 (no activity)
AMT_PAYMENT_CURRENT	20.0%	No payment made OR data gap	Flag as missing_payment=1, fill 0
AMT_INST_MIN_REGULARITY	7.9%	New accounts, no min due yet	Forward-fill or fill 0
CNT_DRAWINGS_* (ATM/POS/Other)	19.5%	Same as drawing amounts	Fill with 0
CNT_INSTALLMENT_MATURE_CUM	7.9%	Same as min regularity	Forward-fill

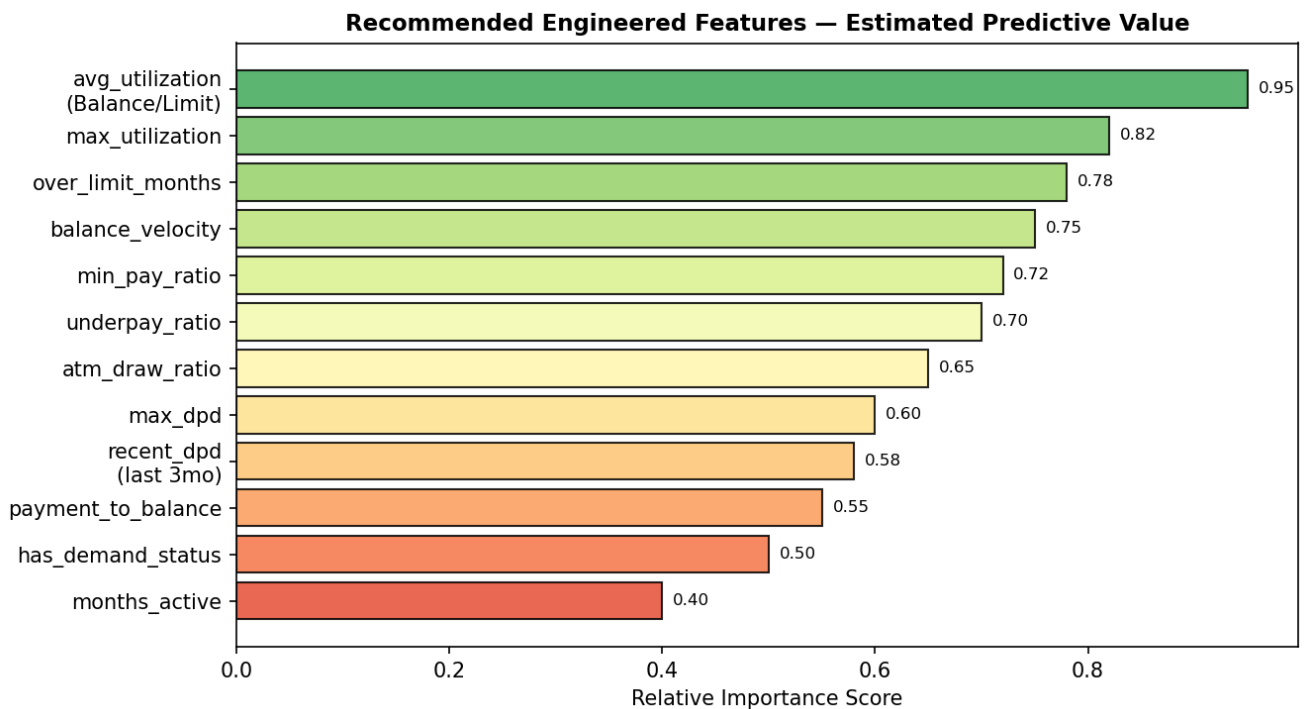
Critical note: The 20% missing in AMT_PAYMENT_CURRENT is potentially the **most dangerous signal** — a missing payment record may indicate the customer didn't pay at all. Models should treat this as an informative missing pattern, not random missingness.

5.2 Data Quality Issues

- Negative balances:** AMT_BALANCE has a minimum of -420,250. This indicates refund/overpayment scenarios or data entry errors. Six features show theoretically impossible negative values.
- Zero credit limits:** Some records have AMT_CREDIT_LIMIT_ACTUAL = 0, causing division-by-zero in utilization calculations. These likely represent closed or frozen accounts.
- Extreme outliers:** Maximum balance (1,505,902) is 25x the mean (58,300). Maximum DPD of 3,260 days (~9 years) is implausible — capping at the 99th percentile is recommended.
- Column naming error:** 'AMT_RECIVABLE' is misspelled (should be 'RECEIVABLE'). Additionally, AMT_RECIVABLE, AMT_TOTAL_RECEIVABLE, and AMT_RECEIVABLE_PRINCIPAL are nearly identical (~0.99 correlation) — two of three can be dropped.
- Temporal survivorship:** Recent months have fewer records than mid-range months, suggesting account closures or portfolio changes over time.

6. Data Scientist View — Predictive Feature Engineering

Based on the analysis, the following features provide the highest predictive value for default risk when aggregated to the applicant level:

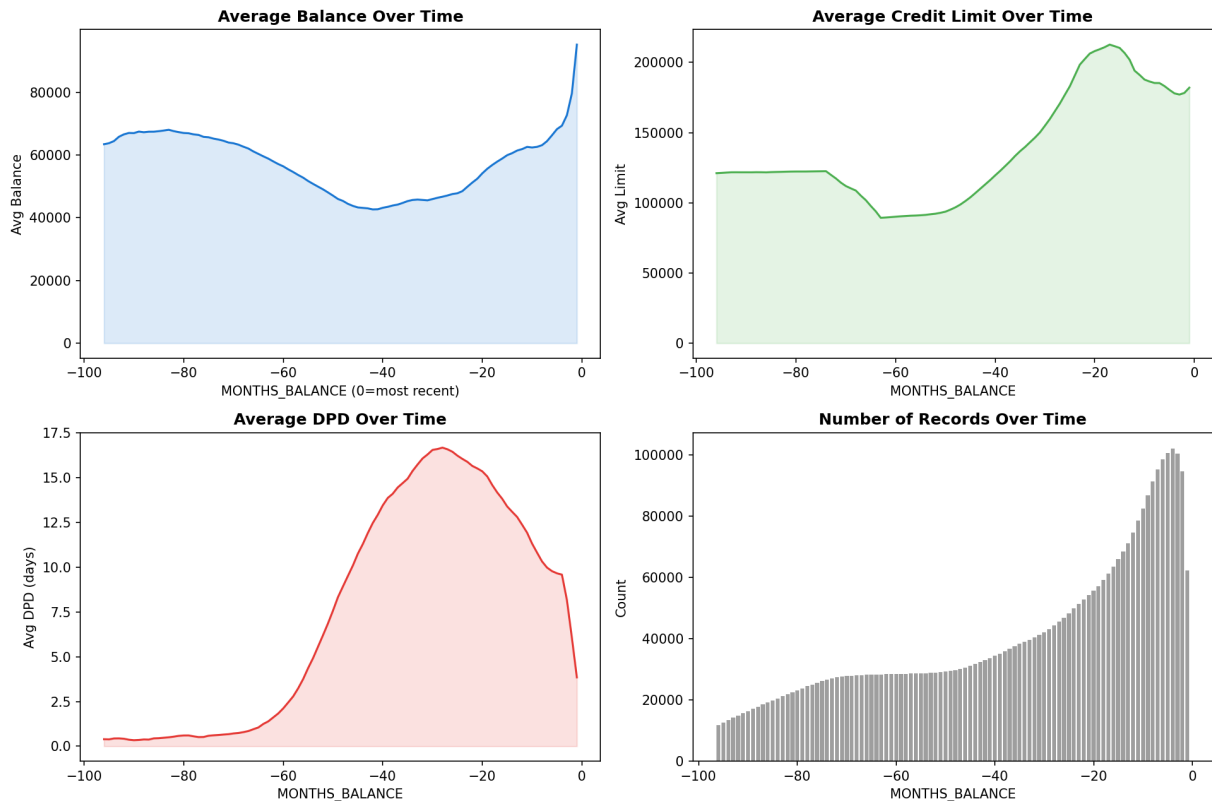


Feature	Logic	Why It Matters
cc_avg_utilization	Mean(balance/limit)	25.5% default at >100% vs 5.6% at <20%
cc_max_utilization	Max utilization ever	Captures peak financial stress moment
cc_over_limit_months	Count(util > 1.0)	Chronic over-limit = persistent risk
cc_balance_velocity	Recent avg - Older avg	Growing debt = deteriorating credit
cc_min_pay_ratio	Min-pay months / total	"Minimum payment trap" indicator
cc_underpay_ratio	Below-min months / total	Direct delinquency precursor
cc_atm_draw_ratio	ATM draws / total draws	Cash-advance-heavy = risk signal
cc_max_dpd	Max DPD reached	Extreme past delinquency (with recency)
cc_recent_dpd	DPD in last 3 months	Current/imminent risk indicator
cc_payment_to_balance	Avg payment / avg bal	Repayment capacity measure
cc_has_demand	Any "Demand" status	Strongest single binary signal
cc_months_active	Count active months	Account maturity / tenure proxy

The inverted-U pattern in payment amounts is notable: both very low and very high payers have lower default rates. Low payers carry small balances; high payers are clearing debt. **Mid-range payers carrying revolving balances are the riskiest segment** (10–11.2% default).

7. Temporal Trend Analysis

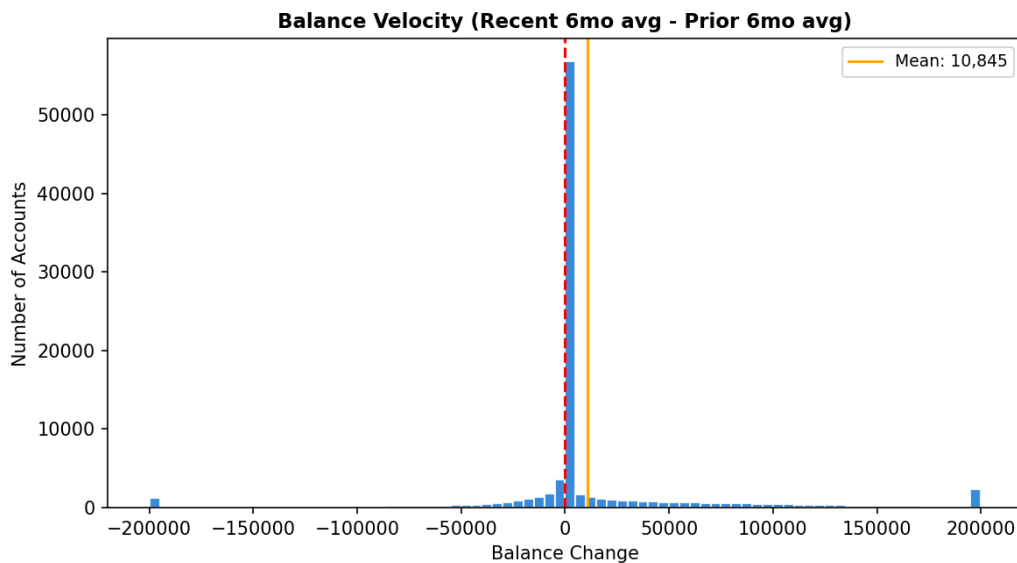
Temporal Trends in Credit Card Portfolio



Average Balance has been rising over the 96-month period, indicating growing credit card debt in the portfolio. The **average credit limit** also grew (from ~121K to ~182K), but balance growth outpaced limit growth — meaning **utilization is increasing over time**.

Average DPD increased significantly in recent months (from ~0.4 at month -96 to ~12.4 at month -1), suggesting either portfolio aging effects or deteriorating credit quality. The **record count** peaks around month -4 to -5 (~102K) before dropping sharply in the most recent months.

7.1 Balance Velocity

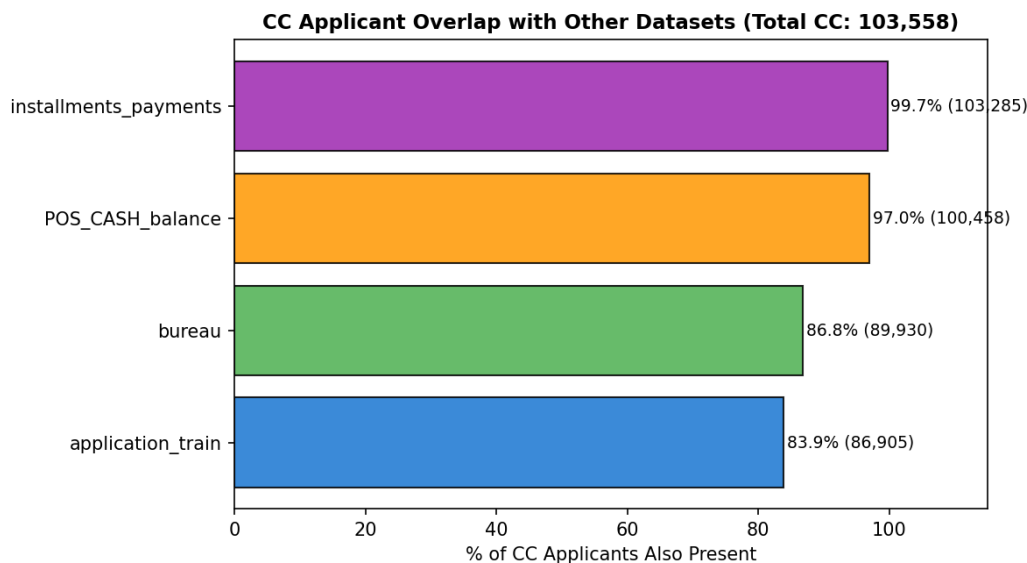


Direction	% of Accounts	Interpretation
Increasing balance	26.3%	Growing debt — risk increasing
Decreasing balance	16.9%	Paying down debt — positive signal
Stable (within +/-10)	57.0%	Steady revolvers or inactive cards

The **mean velocity is +11,940** — on aggregate, the credit card portfolio is experiencing growing indebtedness, which is a macro-level portfolio risk factor.

8. Cross-Dataset Linkage & Value Assessment

8.1 Overlap with Other Tables



Dataset	Overlap with CC	What CC Adds
application_train	86,905 (28.3% of train)	Behavioral dynamics that static features cannot capture
bureau.csv	86.8% of CC holders	Granular utilization — bureau has no usage-level detail
POS_CASH_balance	97.0%	Revolving behavior — POS has only fixed installments
installments_payments	99.7%	Credit limit management & spending channel mix

8.2 Unique Value Proposition of Credit Card Data

The application table has **static** point-in-time features. Credit card balance adds **dynamic behavioral signals**:

- Utilization signal** — the strongest predictor in this table (5.6% vs 25.5% default by band) is **unavailable** in any other dataset.
- Revolving behavior** — POS/Cash data captures fixed installment discipline, but credit cards capture **financial discipline under choice** (how much to pay when there's no fixed schedule).
- Cash advance behavior** — ATM withdrawals from credit cards are a unique risk signal available only here. Bureau data shows credit counts but not usage patterns.
- Payment discipline** — minimum-payment frequency and underpayment ratios reveal financial management quality impossible to infer from the application form.

Combined power example: An applicant with clean bureau history BUT high CC utilization (>80%) defaults at 16.2% — double the baseline. Bureau alone would miss this. Combining bureau credit status + CC utilization creates a 2D risk grid more powerful than either source alone.

8.3 Recommended Cross-Dataset Features

Combined Feature	Tables Used	Purpose
total_credit_exposure	CC + bureau + POS	Total active debt across all products
cc_util × bureau_dpd	CC + bureau	Interaction: high util AND external delinquency

Combined Feature	Tables Used	Purpose
cc_vs_installment_discipline	CC + installments	Compare revolving vs fixed payment behavior
product_diversity_risk	CC + POS + bureau	N product types × max utilization
cc_trend_vs_income	CC + application	Balance velocity / declared income
util × ext_source	CC + application	High util + low EXT_SOURCE = extreme risk

9. Recommended Feature Engineering Pipeline

The following pipeline transforms the raw 3.8M-row panel data into applicant-level features for the main model:

Step	Action	Details
1	Handle missing values	Fill drawing amounts with 0; create missing_payment flag for AMT_PAYMENT_CURRENT
2	Compute utilization	$util = AMT_BALANCE / AMT_CREDIT_LIMIT_ACTUAL$ (handle 0 limits)
3	Payment classification	Below minimum / at minimum / above minimum per month
4	Drawing channel flags	Binary: any_ATM_draw, any_POS_draw per month
5	Aggregate per account	Mean/max/trend for balance, utilization, DPD, payment
6	Aggregate per applicant	Mean/max across accounts (most have 1 account)
7	Temporal features	Balance velocity (6mo window), recent DPD (last 3 months)
8	Status flags	has_demand, has_completed, pct_active_months
9	Cross-table join	Merge on SK_ID_CURR; fill missing with 0 or indicator
10	Feature selection	Remove redundant receivable columns; keep ~12 core features

10. Conclusions & Strategic Recommendations

10.1 Key Takeaways by Perspective

Perspective	Top Insight
Business	Utilization >80% is the strongest default predictor (16–25% vs 5.6%) — should trigger automated limit reviews
Product	25.7% in minimum-payment trap, 4.1% underpaying — proactive restructuring offers could reduce losses
Data Quality	Negative balances, zero limits, extreme DPD outliers (9 years), and survivorship bias require careful handling
Data Science	Top features: avg utilization, balance velocity, min-payment frequency — these belong in every feature set
Cross-Dataset	EC data provides the ONLY source of revolving credit behavior — utilization is unavailable in any other table

10.2 Strategic Recommendations

For Risk Management:

1. Implement real-time utilization monitoring with automated triggers at 70%, 85%, and 95% thresholds. Flag any 'Demand' status transitions for immediate senior review.
2. Build a separate early-warning model that specifically monitors credit card behavioral changes (rising utilization, declining payments, channel shifts) as leading indicators of default.

For Modeling:

3. Always include `cc_avg_utilization` as a feature — it provides a 4.5x default rate spread that no other single feature matches.
4. Handle the DPD survivorship bias by using recent DPD (last 3 months) rather than historical max DPD as the primary delinquency feature.
5. Create cross-table interaction features (`utilization × EXT_SOURCE`, `utilization × bureau_dpd`) to capture multiplicative risk.

For Portfolio Management:

6. The aggregate balance velocity of +11,940 per account indicates growing portfolio indebtedness. Consider tightening credit limits for the 26.3% of accounts with increasing balances.
7. Design targeted interventions for the 25.7% minimum-payment customers: payment plan offers, interest rate reductions for higher payments, or financial counseling referrals.

— End of Report —